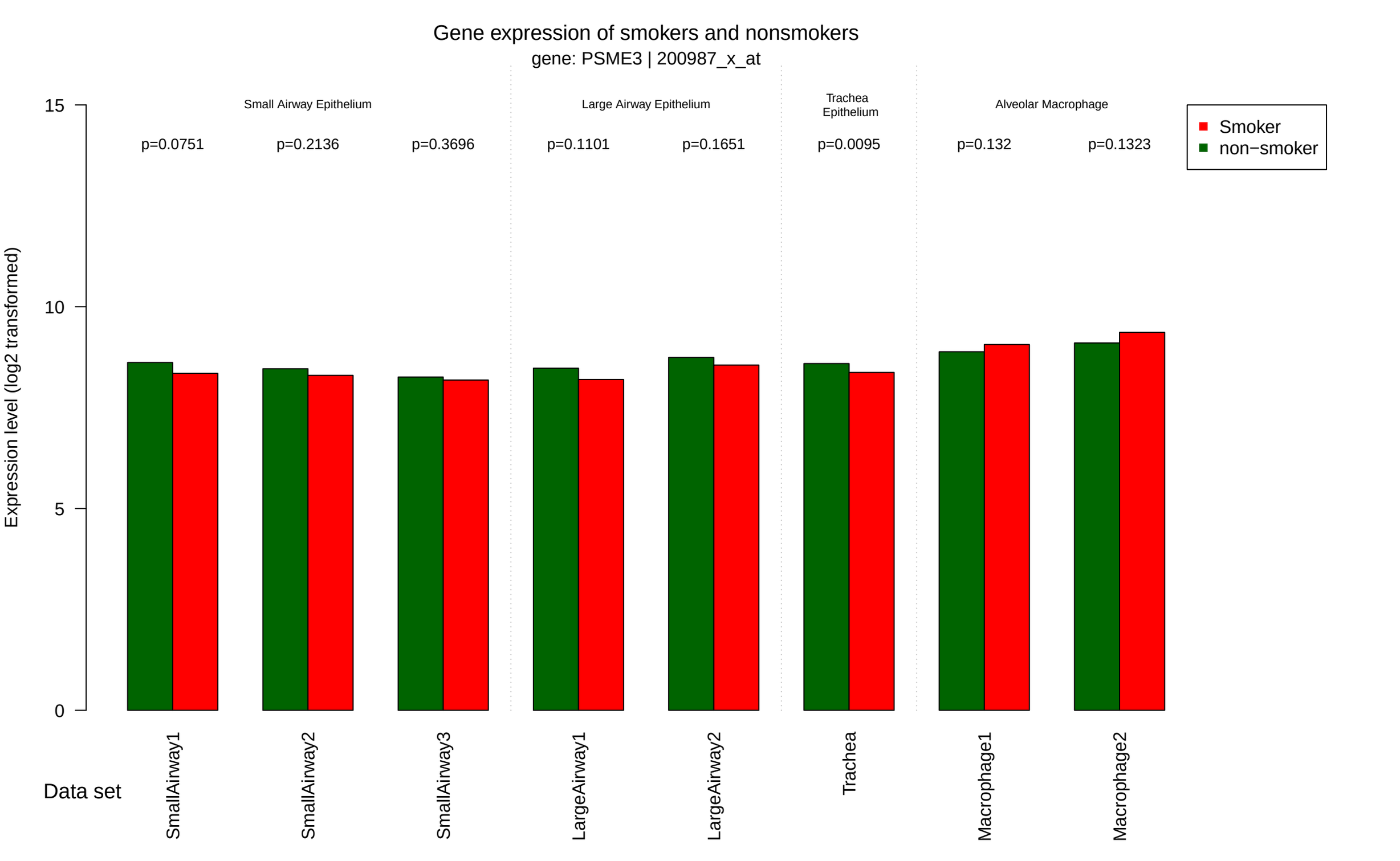
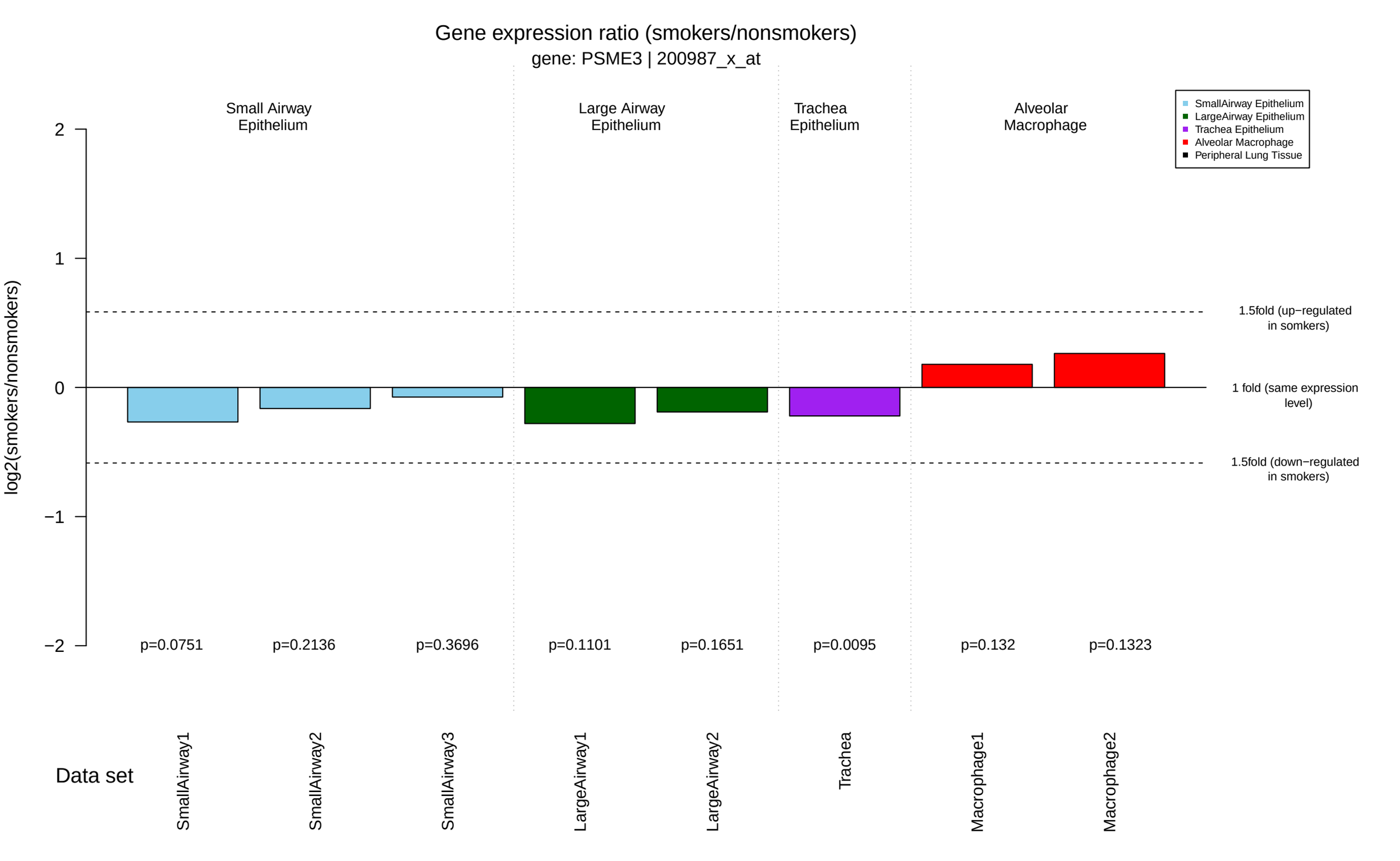
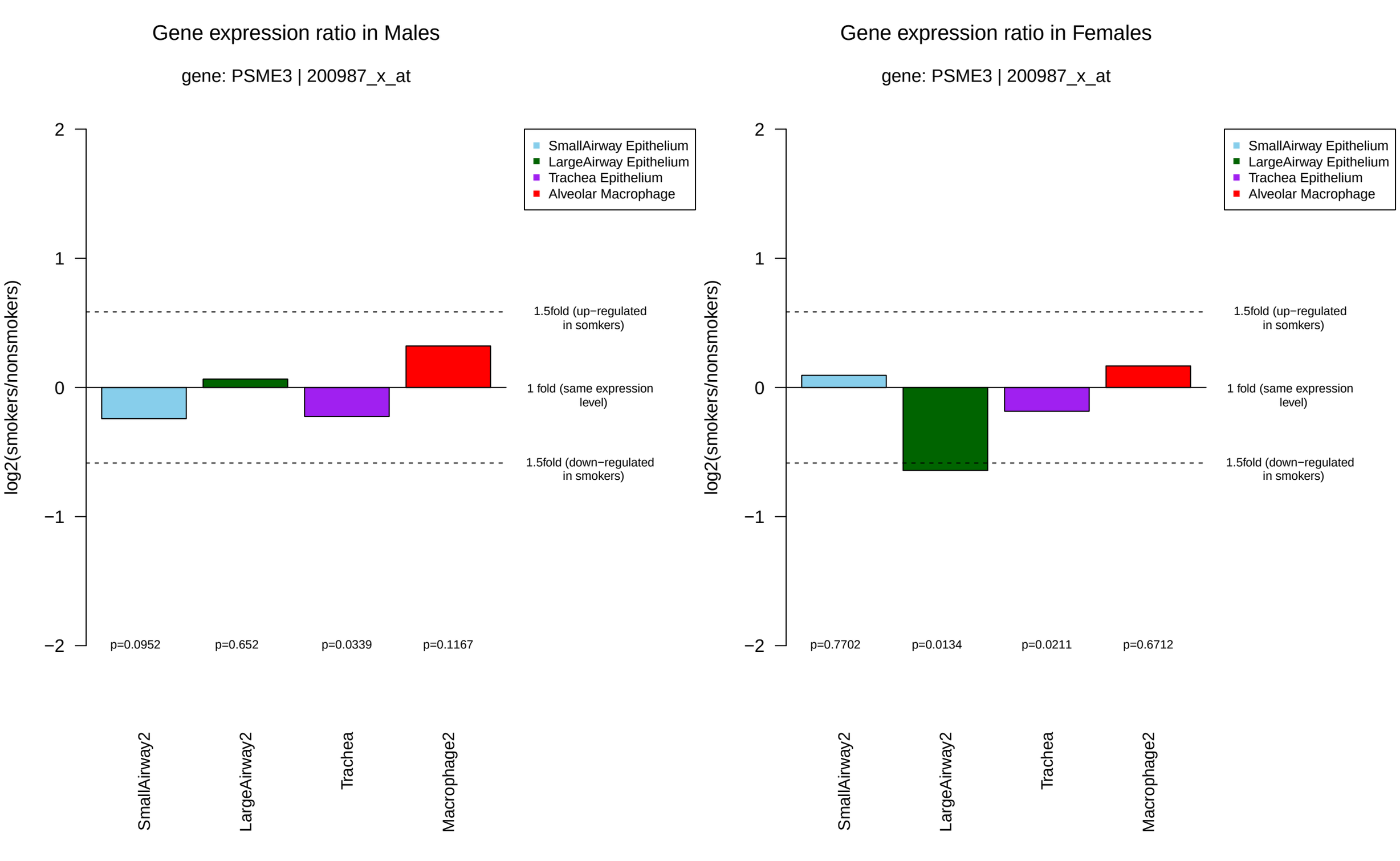
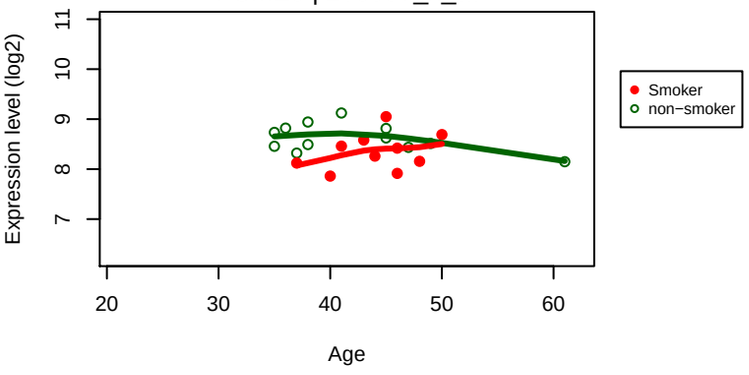
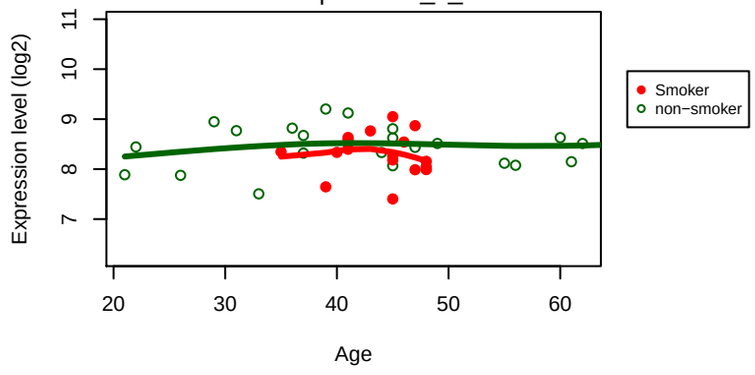
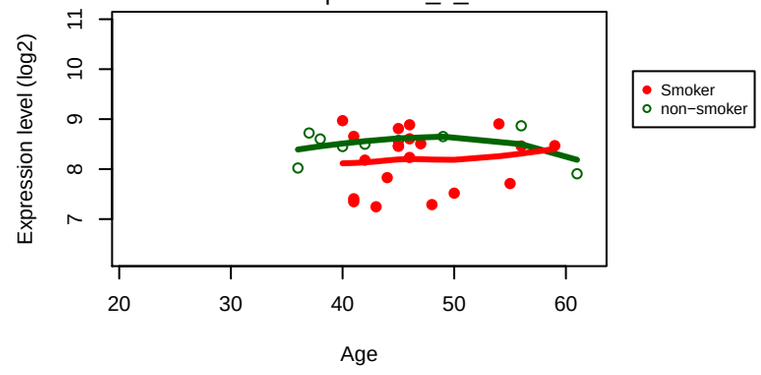

Gene expression of smokers and nonsmokers
gene: PSME3 | 200987_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0751
p=0.2136
p=0.3696
p=0.1101
p=0.1651
p=0.0095
p=0.132
p=0.1323
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

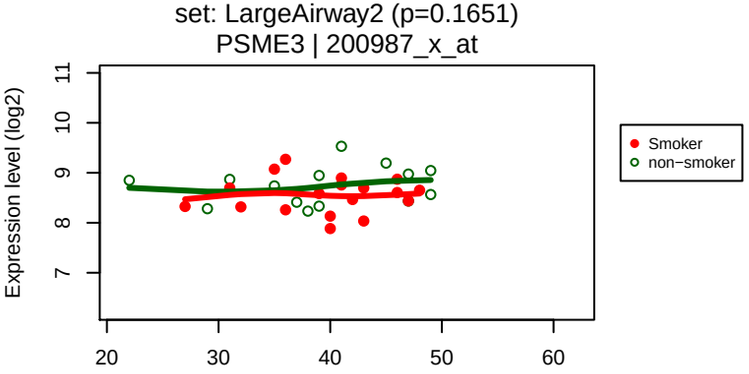

Gene expression ratio (smokers/nonsmokers)
gene: PSME3 | 200987_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0751
p=0.2136
p=0.3696
p=0.1101
p=0.1651
p=0.0095
p=0.132
p=0.1323
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

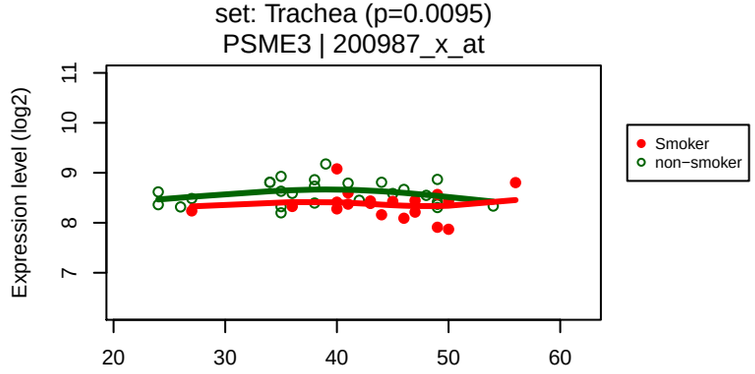

Gene expression ratio in Males
gene: PSME3 | 200987_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0952
p=0.652
p=0.0339
p=0.1167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSME3 | 200987_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7702
p=0.0134
p=0.0211
p=0.6712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

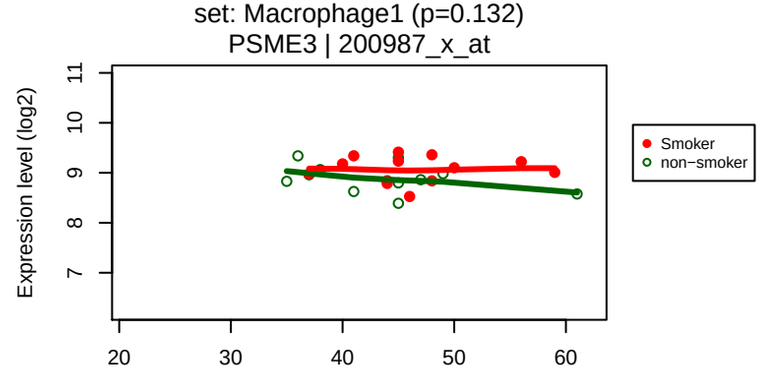

set: SmallAirway1 (p=0.0751)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker

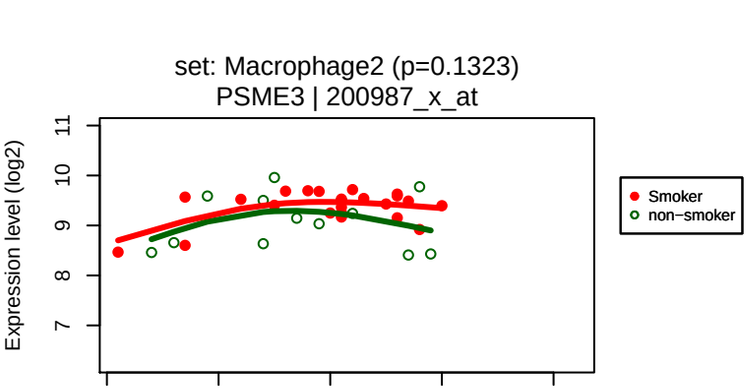

set: SmallAirway2 (p=0.2136)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker

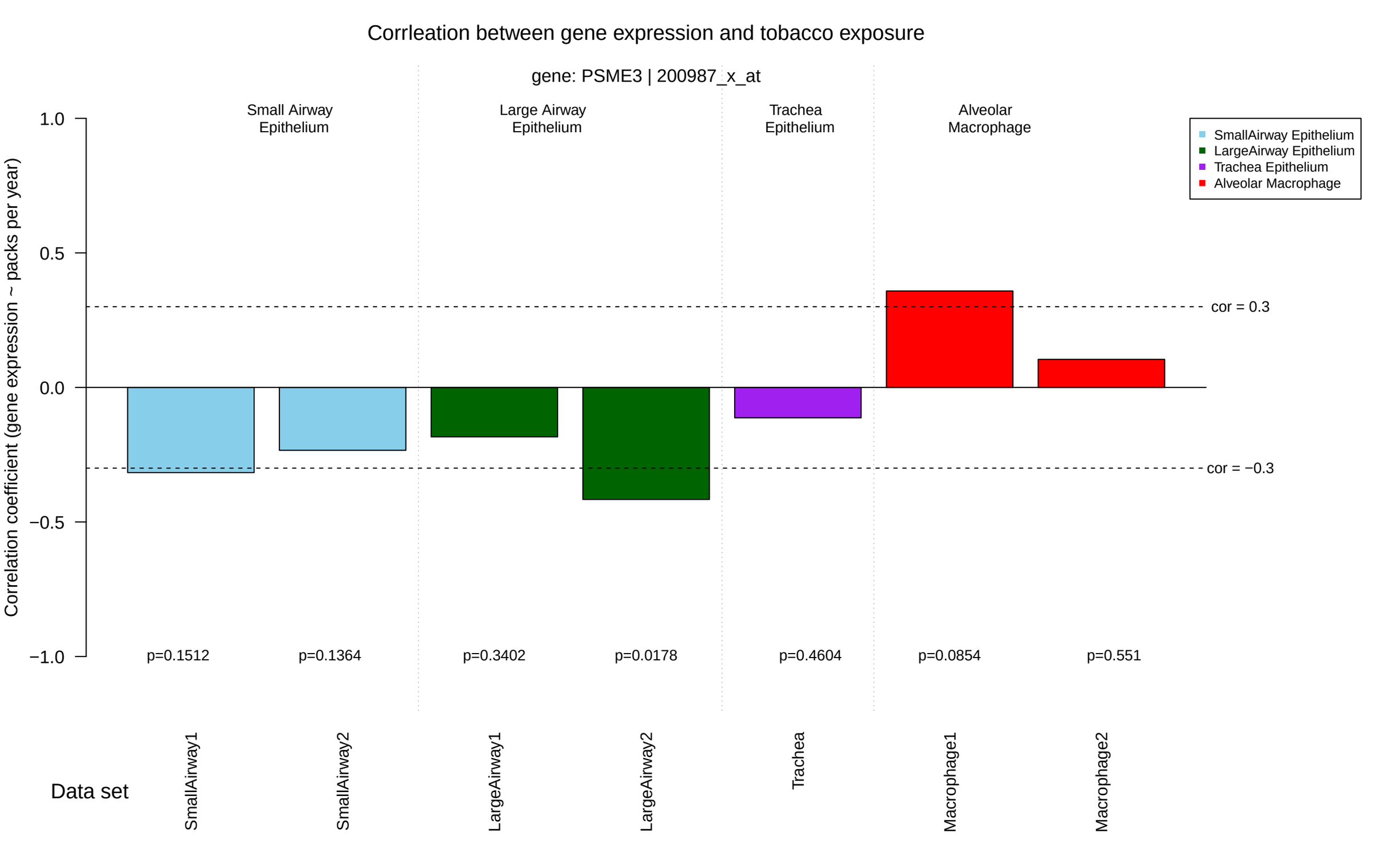

set: LargeAirway1 (p=0.1101)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1651)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.132)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1323)
PSME3 | 200987_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSME3 | 200987_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1512
p=0.1364
p=0.3402
p=0.0178
p=0.4604
p=0.0854
p=0.551
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2